An AI-Enabled Intelligent Study Assistant AuraLearn to Enhance Student Learning Outcomes

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Hallucination- Controlled LLM-Based Tutor

Project Proposal Report

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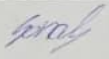
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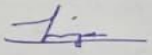
DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

University students increasingly seeking help from AI-based study tools for academic support, yet current Large Language Models (LLMs) tutors sometimes generate hallucinated or unreliable explanations, especially when answering to queries that are outside their training data distribution. Such mistakes cause significant risks in educational contexts, where misleading information negatively impact towards learning outcomes, exam preparation, and student confidence and mentality. Currently existing Retrieval-Augmented generation (RAG) systems partially address this issue by grounding responses in external documents, but they mostly use static prompting strategies and currently lacks the mechanism to evaluate the appropriateness or reliability of generated answers. This creates a critical research gap in developing LLM-based tutoring systems that can dynamically assess uncertainty and adapt their response style to minimize hallucinations. This research proposes a confidence-aware, hallucination-controlled LLM tutoring system grounded in course-specific academic materials. The solution integrates a vector-database-driven RAG pipeline with uncertainty estimation techniques to evaluate model confidence before generating a response. Based on confidence levels, the system intelligently selects among three adaptive strategies which are providing a direct answer, offering a guided hint, or requesting clarification from the student. This approach ensures that explanation remains comparatively accurate, contextually grounded, verifiable and pedagogically appropriate. Course materials such as lecture slides, textbooks, and learning outcomes are indexed to guarantee that all responses originate from verified academic sources. The system will be evaluated using hallucination rate, answer accuracy, and student feedback to measure reliability and educational usefulness. Expected outcomes include a significant reduction in hallucinated responses, improved trust in AI-based tutoring, and enhanced learning effectiveness. By introducing confidence-aware response control into LLM tutoring, this research contributes a novel and practical advancement to intelligent educational systems.

Keywords: Large Language Models, RAG, Hallucination Reduction, Confidence-Aware AI, Intelligent Tutoring Systems

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
NLP	Natural Language Processing
DB	Database
API	Application Programming Interface
LMS	Learning Management System
SDG	Sustainable Development Goal
DSR	Design Science Research
SaaS	Software as a Service
MOOC	Massive Open Online Course
S3	Amazon Simple Storage Service
FAISS	Facebook AI Similarity Search

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1. INTRODUCTION

1.1 Background and Context

Artificial Intelligence–driven study tools have rapidly transformed modern higher education, offering students new ways to understand complex concepts, receive personalized explanations, and prepare for examinations. Recent advancements in Large Language Models (LLMs) such as GPT-based systems have demonstrated strong capabilities in generating explanations, solving problems, and supporting self-directed learning. Studies [1] show that multimodal AI tutors can enhance affective engagement and conceptual understanding by combining text, visuals, and emotional cues. Similarly, multimodal tutoring frameworks in engineering education have demonstrated improved personalization and learner satisfaction [2]. These developments highlight the growing potential of AI-powered tutoring systems to support diverse learning needs.

Despite these advancements, LLMs still suffer from a critical limitation: hallucinations, where the model generates factually incorrect or unsupported information. [5] emphasize that LLM-powered tutoring systems often produce inaccurate explanations when lacking contextual grounding, which can mislead students and hinder deep learning. Even systems that incorporate Retrieval-Augmented Generation (RAG), such as [4], reduce hallucinations only partially because they rely on static prompting and do not evaluate the reliability of generated responses. This issue is especially problematic in academic environments where accuracy is essential.

Recent AI tutoring platforms—including [3], [7], and [9] demonstrate the growing trend toward adaptive, personalized learning support. However, these systems still lack mechanisms to verify whether generated answers are grounded in course-specific materials. They also do not incorporate uncertainty estimation, meaning they provide direct answers even when the model is unsure. This limitation increases the risk of hallucinations, particularly during high-stakes learning scenarios such as exam preparation.

Furthermore, research in intelligent tutoring emphasizes the importance of adaptivity and learner engagement. Adaptive generative AI tutors [6] and earlier conversational support systems [8] highlight the need for systems that adjust their behavior based on learner context. Engagement detection studies [11] further show that effective tutoring requires dynamic responses aligned with learner needs. However, current LLM-based tutors do not incorporate confidence-aware adaptivity, leaving a significant gap in reliability and pedagogical alignment.

In summary, while AI tutoring systems have advanced significantly, existing solutions still lack confidence awareness, dynamic response control, and course-specific grounding. These limitations create a critical need for a hallucination-controlled, confidence-aware LLM tutoring system capable of delivering trustworthy, academically aligned explanations.

1.2 Literature Review

Recent studies have explored the integration of Large Language Models (LLMs) into educational tutoring systems, highlighting both their strengths and persistent limitations. Jiang and Jiang (2024) demonstrate that LLM-powered tutors can significantly enhance conceptual understanding in subjects such as physics; however, they also show that these systems frequently generate incorrect or unverified explanations when they lack contextual grounding. This issue becomes more severe in academic settings where factual accuracy is essential for effective learning.

To address hallucinations, researchers have investigated Retrieval-Augmented Generation (RAG) pipelines. [4] Through their LPITutor system, show that RAG can reduce hallucinations by retrieving relevant documents before generating responses. Yet, their findings also reveal that existing RAG-based tutors rely on static prompting and do not evaluate the reliability or appropriateness of the generated answers. As a result, these systems still produce misleading explanations when retrieval is weak or when the model is uncertain.

Beyond hallucination reduction, recent LLM-based tutoring systems such as MyLearningTalk [7] focus on conversational learning support but lack mechanisms to verify whether responses are grounded in course-specific materials. These systems also do not incorporate uncertainty estimation, meaning they provide direct answers even when confidence is low. This makes them unsuitable for high-stakes academic environments where students depend on trustworthy explanations.

Adaptive tutoring systems powered by generative AI, such as the framework proposed by [6] highlight the importance of adjusting explanations based on learner needs. However, these systems still do not integrate confidence-aware decision-making or dynamic response control. They adapt content style but not response reliability, leaving a gap in ensuring factual correctness.

Overall, the existing literature highlights a clear need for LLM tutoring systems that combine course-specific grounding, uncertainty estimation, and adaptive response strategies. Current systems either reduce hallucinations partially or improve personalization, but none integrate confidence-aware control to ensure reliable, academically aligned explanations.

1.3 Research Gap

Although recent advancements in LLM-based tutoring systems have improved personalization and content generation, several critical limitations remain unresolved. Studies [5] shows that LLM tutors frequently produce incorrect or unverified explanations when lacking contextual grounding, making them unreliable in academic environments. RAG-based systems like LPITutor [4] partially address this issue by retrieving relevant documents before generating responses, but they still rely on static prompting and do not assess the reliability or confidence of the generated output. As a result, these systems continue to produce hallucinations when retrieval is weak or ambiguous.

Furthermore, emerging adaptive generative AI tutors Almetnawy [6] and LLM tutoring platforms such as MyLearningTalk [7] focus on conversational support and personalized content delivery but do not incorporate uncertainty estimation or dynamic response control. These systems adapt what they say but not whether they should answer at all, leading to situations where low-confidence responses are still presented as factual. None of the existing solutions integrate confidence-aware decision-making with course-specific grounding to ensure academic reliability.

Therefore, the research gap lies in the absence of a confidence-aware, hallucination-controlled LLM tutoring system that:

- Grounds responses strictly in verified academic materials
- Evaluates model uncertainty before generating an answer
- Adapts its response strategy (direct answer, guided hint, or clarification request) based on confidence
- Ensures reliability, transparency, and trustworthiness in educational settings

This research addresses the gap by integrating uncertainty estimation with a RAG pipeline and adaptive tutoring logic, enabling the system to avoid hallucinations and deliver academically aligned, trustworthy explanations and advancement not present in current academic or commercial tutoring systems.

2. OBJECTIVES

2.1 Main Objective

To design and implement a hallucination-controlled, confidence-aware LLM-based tutoring system that generates course-grounded academic responses by integrating a Retrieval-Augmented Generation (RAG) pipeline with uncertainty-based adaptive response strategies, and to achieve a measurable reduction in hallucination rate and improvement in answer accuracy during evaluation.

2.2 Specific Objectives

This

1. Development of a Confidence-Aware Retrieval-Augmented Generation (RAG) Pipeline

- Identify and evaluate suitable embedding models (e.g., Sentence-BERT, OpenAI embeddings) for indexing course-specific academic materials such as lecture slides, textbooks, and learning outcomes.
- Design and implement a vector-database-driven retrieval pipeline capable of retrieving semantically relevant academic content with high accuracy.
- Integrate uncertainty estimation techniques—including log-probability scoring, entropy analysis, and self-consistency checks—to quantify model confidence prior to response generation.
- Validate the RAG pipeline across a diverse set of academic queries to assess retrieval precision, grounding quality, and robustness.

2. Implementation of Adaptive, Hallucination-Controlled Response Strategies

- Develop a multi-tier adaptive response mechanism that selects between direct answers, guided hints, or clarification prompts based on predefined confidence thresholds.
- Design pedagogically aligned hint-generation logic to support student learning without revealing full solutions prematurely.
- Implement fallback strategies for low-confidence scenarios to prevent hallucinated or unsupported responses.
- Evaluate the effectiveness of adaptive responses through controlled experiments and student feedback to measure improvements in clarity, trust, and learning outcomes.

3. System Evaluation, Reliability Testing, and Ethical Assurance

- Define measurable evaluation metrics such as hallucination rate, answer accuracy, grounding score, and response appropriateness.
- Conduct comparative testing between grounded and ungrounded LLM responses to quantify the impact of confidence-aware control mechanisms.

- Perform user-based evaluations with students to assess perceived usefulness, trustworthiness, and clarity of the tutoring system.
- Ensure ethical AI behavior by enforcing strict grounding rules, preventing unsupported claims, and maintaining transparency in low-confidence situations.
- Optimize computational efficiency to ensure the system remains scalable, responsive, and deployable within real educational environments.

3. METHODOLOGY

3.1 Research Approach

This research adopts the Design Science Research (DSR) methodology, as the primary objective is to design, develop, and evaluate a novel AI-based tutoring artifact that addresses the problem of hallucinations in LLM-powered academic support systems. DSR is appropriate because it emphasizes the creation of innovative, practical solutions to real-world problems through iterative cycles of design, implementation, and evaluation.

The research process begins with problem identification, informed by prior studies showing that LLM tutors often generate incorrect or unverified explanations when lacking contextual grounding. Following this, the system is designed using a confidence-aware Retrieval-Augmented Generation (RAG) architecture inspired by recent advancements in LLM tutoring systems. A functional prototype is then developed, integrating uncertainty estimation and adaptive response strategies to minimize hallucinations.

The evaluation phase incorporates experimental testing, where the system's hallucination rate, grounding accuracy, and response appropriateness are measured across diverse academic queries. User-based evaluations are also conducted to assess clarity, trustworthiness, and perceived usefulness. Insights from these evaluations feed into iterative refinement cycles, ensuring that the final artifact is reliable, pedagogically aligned, and suitable for deployment in real educational environments.

3.2 Technical Methods and Algorithms

1. Retrieval-Augmented Generation (RAG) Pipeline

To ensure that all responses are grounded in verified academic materials, the system employs a vector-database-driven RAG pipeline:

- Embedding Models: Course materials (lecture slides, textbooks, learning outcomes) are converted into dense vector representations using models such as Sentence-BERT or OpenAI embeddings.
- Vector Search: A semantic similarity search engine (FAISS or ChromaDB) retrieves the most relevant academic content for each student query.
- Document Ranking: Retrieved documents are ranked using top-k relevance scoring to ensure that only the most contextually appropriate materials are provided to the LLM.
- This pipeline ensures that the LLM does not rely solely on internal knowledge, reducing the likelihood of hallucinations.

2. Uncertainty Estimation Algorithms

To detect potential hallucinations, the system incorporates multiple uncertainty estimation techniques:

- Token-level log-probability scoring to measure how confidently the model predicts each token.
- Entropy-based uncertainty measurement to quantify prediction variability.
- Self-consistency checks, where multiple sampled outputs are compared for agreement.
- Confidence thresholds that classify responses into high, medium, or low certainty categories.
- These mechanisms allow the system to evaluate the reliability of its own output before responding to the student.

3. Adaptive Response Strategy
Based on the computed confidence score, the system dynamically selects one of three pedagogically aligned response modes:

- High confidence → Direct answer
- Medium confidence → Guided hint
- Low confidence → Clarification request
- This adaptive strategy ensures that the system avoids providing incorrect explanations and instead supports learning through safe, context-aware interactions.

4. Grounding Enforcement
To maintain academic reliability:

- All responses must reference retrieved course materials.
- If grounding is weak or retrieval confidence is low, the system automatically switches to hint or clarification mode.
- This prevents unsupported claims and ensures that every explanation is traceable to verified academic source

3.3 Tools and Platforms (with Justification)

Tool/Platform	Purpose	Justification
Python + FastAPI	Backend development and API orchestration	Lightweight, high-performance framework widely used for AI-driven systems and rapid prototyping
Hugging Face Transformers	LLMs and embedding models	Provides state-of-the-art NLP models essential for RAG, embedding generation, and uncertainty analysis
FAISS / ChromaDB	Vector database for semantic retrieval	Enables fast, scalable similarity search required for efficient RAG pipelines
PyTorch / TensorFlow	Model fine-tuning & inference	Industry-standard deep learning frameworks with strong support for LLM experimentation

Tool/Platform	Purpose	Justification
Jupyter Notebook	Experimentation and evaluation	Ideal for testing algorithms, analyzing uncertainty metrics, and validating retrieval performance
Postman	API testing	Ensures correctness, reliability, and stability of backend endpoints
GitHub	Version control and collaboration	Supports reproducibility, team collaboration, and continuous integration

Table. 1-Tools and Platforms of component Hallucination – controlled, confidence-Aware LLM Tutoring Module

3.4 Data Collection and Ethical Considerations

Data Collection

- Course materials—including lecture slides, textbooks, tutorial sheets, and learning outcomes—will be collected from university provided or publicly accessible academic sources.
- These documents will be preprocessed and converted into vector embeddings for use within the RAG pipeline.
- No personal or sensitive student data will be collected, stored, or processed at any stage of the research.
- All data used in the system is strictly academic and non identifiable, ensuring compliance with institutional ethical guidelines.

Ethical Considerations

- The system will not process or store sensitive information such as student identities, behavioral data, or webcam inputs.
- All generated responses must be grounded in verified academic content to prevent misinformation and hallucinated explanations.
- The system will maintain transparency by indicating low confidence responses and avoiding unsupported claims.
- Data will be stored securely and used solely for research and evaluation purposes.
- Ethical AI principles—such as fairness, reliability, and accountability—will be upheld throughout system development and testing.

3.5 Validation Strategy and Performance Metrics

Validation Strategy

1. Technical Evaluation

- Compare grounded (RAG-enabled) responses with ungrounded LLM responses to measure improvements in accuracy and reliability.
- Test system behavior under high-confidence and low-confidence scenarios to evaluate the effectiveness of uncertainty estimation.
- Assess retrieval relevance, grounding quality, and response appropriateness across diverse academic queries.

2. User Evaluation

- Conduct small-scale evaluations with university students to assess clarity, usefulness, and trustworthiness of the system.
- Collect qualitative and quantitative feedback on the effectiveness of direct answers, hints, and clarification prompts.

3. A/B Testing

- Compare standard LLM responses with confidence-aware adaptive responses.
- Measure the reduction in hallucination rate and improvement in perceived reliability.

Performance Metrics

- Hallucination Rate (%) -
  Percentage of responses containing incorrect, fabricated, or unsupported information.
- Answer Accuracy (%) -
  Degree of alignment between generated responses and verified academic materials.
- Grounding Score -
  Extent to which responses correctly cite or rely on retrieved course documents.
- Confidence–Response Alignment -
  Whether the system correctly selects direct answers, hints, or clarification requests based on uncertainty thresholds.
- User Satisfaction Score -
  Measured through Likert-scale surveys evaluating clarity, trust, and usefulness.
- Retrieval Precision & Recall -
  Effectiveness of the RAG pipeline in retrieving relevant academic content.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed system is a hallucination-controlled, confidence-aware LLM tutoring module designed to generate accurate, course-grounded academic explanations. Although it functions as a core component of the broader Smart Study Assistant, it operates as an independent, self-contained subsystem. The architecture integrates a Retrieval-Augmented Generation (RAG) pipeline, uncertainty estimation mechanisms, and adaptive response logic to ensure that all generated outputs are reliable, pedagogically aligned, and grounded in verified academic materials.

The system processes each student query through a structured sequence of stages:

1. **Query Understanding**
2. **Document Retrieval from Vector Database**
3. **Confidence Estimation**
4. **Adaptive Response Selection**
5. **Grounded Output Delivery**

This architecture ensures that the tutor does not rely solely on the LLM's internal knowledge but instead uses course-specific materials as the authoritative source of truth, significantly reducing hallucinations and improving academic reliability.

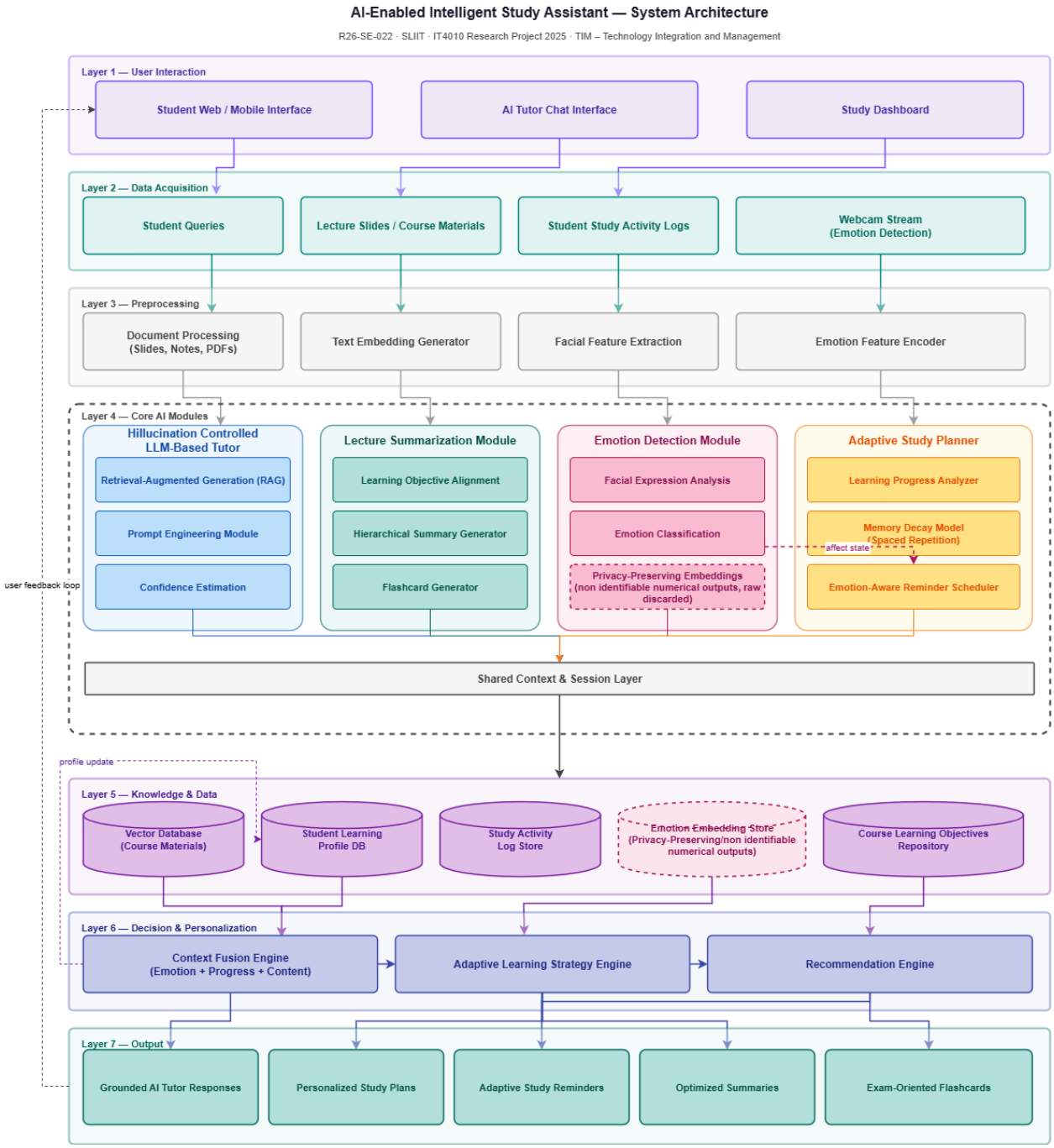


Figure 0-1-An AI-Enabled Intelligent Study Assistant AuraLearn to Enhance Student Learning Outcomes

4.2 System Components

1. Query Processing Layer

- Receives the student's question through the user interface.
- Performs text normalization, tokenization, and intent detection.
- Ensures the query is properly formatted for retrieval and LLM processing.

2. Vector Database & Retrieval Layer

- Stores embeddings of lecture slides, textbooks, learning outcomes, and tutorial notes.
- Uses semantic similarity search (FAISS/ChromaDB) to retrieve the most relevant documents.
- Ensures grounding by supplying the RAG engine with authoritative academic context.

3. Confidence Aware RAG Engine

- Combines retrieved documents with the student query.
- Generates candidate responses using the LLM.
- Computes uncertainty using:
- Log probability scoring
- Entropy based uncertainty measurement
- Self consistency checks
- Produces a confidence score for each generated response.

4. Adaptive Response Controller

- Applies confidence thresholds to determine the appropriate response mode:
    - High confidence → Direct answer
    - Medium confidence → Guided hint
    - Low confidence → Clarification request
- Prevents hallucinations by avoiding direct answers when confidence is low.
- Ensures pedagogical appropriateness and safe learning support.

5. Grounded Output Generator

- Formats the final response for delivery.
- Ensures citations or references to retrieved course materials.
- Removes unsupported claims or hallucinated content.
- Sends the final output to the user interface.

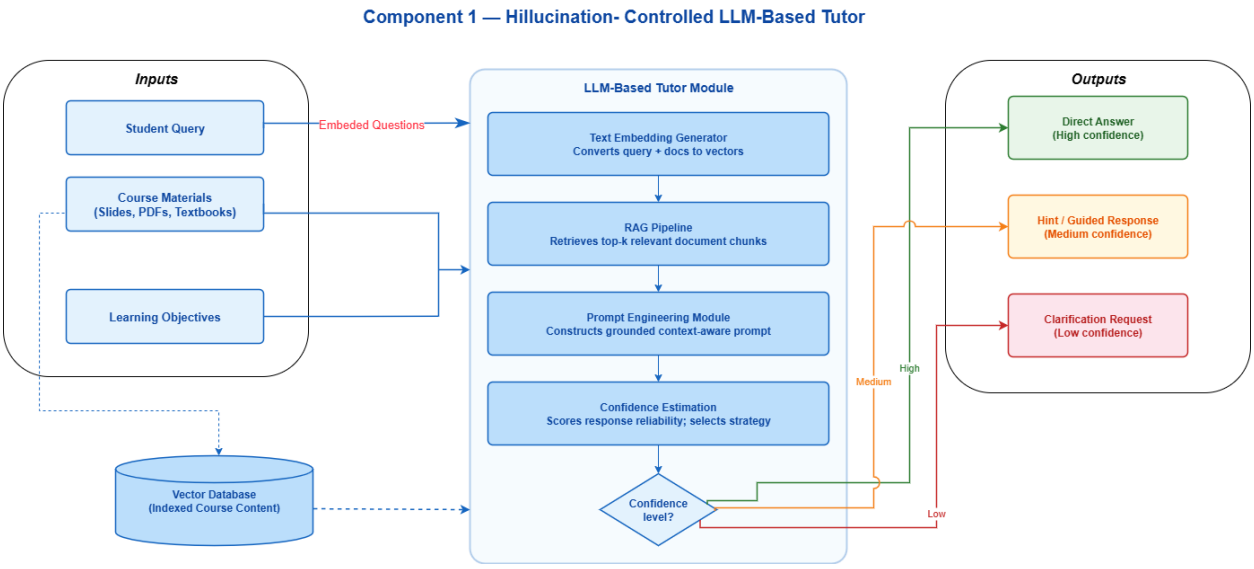


Figure 0-2-Component Architecture of the Confidence Aware RAG Tutor

4.3 Component Interactions

- The student submits a query.
    1. The Query Processing Layer interprets and forwards it to the Retrieval Layer.
    2. The Vector Database returns the top k relevant academic documents.
    3. The RAG Engine generates a response and computes confidence.
    4. The Adaptive Controller selects the appropriate response strategy.
    5. The Grounded Output Generator formats and delivers the final answer.

This interaction ensures that every response is:

- Grounded
- Confidence validated
- Pedagogically aligned
- Free from hallucinations

4.4 Data Flow Explanation

The system follows a structured data flow that begins with the ingestion of academic materials and ends with the delivery of a grounded, confidence validated response.

1. Academic Material Upload Flow
    - Course instructors or authorized users upload lecture slides, reference books, tutorial sheets, and learning outcomes into the system.
    - These documents are preprocessed, converted into embeddings, and stored in the vector database.
    - This ensures that all future responses are grounded in verified academic content.
2. Input Flow
    - Student query → Query preprocessing → Embedding search → Retrieval of top k academic documents
3. Processing Flow
    - Retrieved documents + Student query → LLM generation → Uncertainty estimation (log probability, entropy, self consistency)

4. Decision Flow
    - Confidence score → Selection of response mode (answer / hint / clarification) → Grounding enforcement
5. Output Flow
    - Final grounded response → Formatting → Delivery to user interface
    - This data flow ensures that the system always prioritizes accuracy, grounding, and reliability, and that no response is generated without referencing course verified materials.

4.5 Scalability and Security Considerations

Scalability

- The vector database supports horizontal scaling to accommodate large repositories of uploaded course materials.
- LLM inference can be optimized using batching, quantization, or model distillation to support high query volumes.
- The modular architecture allows seamless integration into larger educational platforms or LMS systems.
- Preprocessing pipelines can be parallelized to handle bulk uploads of lecture slides and textbooks efficiently.

Security

- Only academic materials (slides, textbooks, notes) are uploaded; no personal student data is collected or stored.
- Uploaded documents are stored securely with restricted access to prevent unauthorized use.

- User queries are processed transiently and not stored long term, ensuring privacy.
- Grounding enforcement prevents the generation of fabricated or harmful content by ensuring all responses reference retrieved academic materials.
- The system maintains transparency by indicating low confidence responses and avoiding unsupported claims.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholder	Role	Interest / Responsibility
University Students	Primary users	Use the LLM tutor to ask academic questions and receive grounded, accurate explanations.
Lecturers / Academic Staff	Content providers	Provide course materials (slides, notes, learning outcomes) used to ground the system.
System Developers / Research Team	Designers & implementers	Build, test, and maintain the hallucination-controlled tutoring module.
Supervisors/Evaluators	Quality assurance	Ensure the system meets academic, ethical oversight, and technical standards.
University IT Department	Deployment support	Provide infrastructure support for hosting and integration.

Table. 2 Stakeholder Identification Table of component Hallucination – controlled, confidence-Aware LLM Tutoring Module

5.2 Functional Requirements

FR1 – Course Material Upload and Preprocessing

- The system shall allow lecturers or authorized staff to upload course materials (lecture slides, textbooks, tutorial sheets, learning outcomes).
- The system shall preprocess uploaded documents and convert them into vector embeddings for storage in the vector database.
- The system shall validate uploaded files to ensure they contain academic content only.

FR2 – Query Processing

- The system shall accept natural language academic questions from students.
- The system shall preprocess the query through normalization, tokenization, and intent detection.
- The system shall ensure the query is formatted correctly for retrieval and LLM processing.

FR3 – Course Material Retrieval

- The system shall retrieve relevant academic documents from the vector database using semantic similarity search.
- The system shall rank retrieved documents based on relevance to the student query.
- The system shall ensure that only course-verified materials are used for grounding.

FR4 – Confidence-Aware Response Generation

- The system shall generate candidate responses using the LLM combined with retrieved documents.
- The system shall compute a confidence score using uncertainty estimation techniques (log-probability, entropy, self-consistency).
- The system shall classify confidence levels into high, medium, or low categories.

FR5 – Adaptive Response Strategy

- The system shall provide a direct answer when confidence is high.
- The system shall provide a guided hint when confidence is medium.
- The system shall request clarification when confidence is low.
- The system shall avoid generating unsupported or hallucinated claims under all conditions.

FR6 – Grounded Output Delivery

- The system shall ensure all responses are grounded in retrieved academic materials.
- The system shall cite or reference the source materials used in generating the answer.
- The system shall format the final response clearly and concisely for the user.

FR7 – Logging and Monitoring

- The system shall log queries, confidence scores, retrieval results, and response types for evaluation.
- The system shall allow developers to review hallucination cases for system improvement.
- The system shall maintain logs securely without storing personal student data.

5.3 Non-Functional Requirements

A. Performance Requirements

- The system shall generate responses within 3–5 seconds under normal load.
- The vector database shall retrieve relevant documents with high precision and recall.
- The system shall maintain consistent performance for multiple concurrent users.
- Preprocessing of uploaded course materials shall complete within acceptable time limits.

B. Accuracy & Reliability Requirements

- The system shall maintain a hallucination rate of less than 10% during evaluation.
- The system shall ensure that 100% of direct answers are grounded in course materials.
- The system shall activate fallback strategies (hints/clarifications) when confidence is low.
- The system shall ensure retrieval relevance through continuous monitoring and evaluation.

C. Usability Requirements

- The system shall provide clear, readable, and student-friendly explanations.
- The interface shall support simple, conversational input.
- The system shall display transparency indicators when confidence is low.
- The system shall provide intuitive navigation for uploading and managing course materials.

D. Security Requirements

- The system shall not store personal student data.
- All uploaded course materials shall be stored securely in the vector database.
- The system shall prevent unauthorized access to academic content.
- All logs shall exclude personally identifiable information

E. Scalability Requirements

- The system shall support scaling of the vector database as more course materials are added.
- The architecture shall allow integration into larger educational platforms or LMS systems.
- The system shall support horizontal scaling for retrieval and inference workloads.

F. Ethical Requirements

- The system shall avoid generating harmful, biased, or fabricated content.
- The system shall maintain transparency by indicating uncertainty when applicable.
- The system shall comply with responsible AI guidelines and academic integrity standards.
- The system shall ensure that all responses are grounded in verified academic sources.

6. Commercialization Plan

6.1 Target Market and Customer Persona

Target Market

- Higher Education Institutions: Universities and institutes seeking reliable, academically grounded AI tutoring solutions.
- EdTech Companies: Platforms integrating intelligent tutoring, adaptive learning, or AI driven study support.
- Online Learning Platforms (MOOCs): Digital academies requiring accurate, verifiable explanations for large student cohorts.
- Corporate Training Providers: Organizations offering technical upskilling programs that require trustworthy AI assistance.
-

Customer Persona

Name: Nethmi – 20 year old undergraduate student
Goals: Understand complex topics, prepare for exams, and receive reliable academic explanations

Pain Points:

- AI tools generating hallucinated or incorrect answers
- Difficulty verifying information
- Lack of personalized academic guidance

How the System Helps:
The proposed tutor eliminates misinformation by providing confidence aware, course grounded responses, ensuring students receive trustworthy explanations.

6.2 Value Proposition

Core Value -
A highly reliable AI tutor that delivers verified, confidence aware academic explanations, minimizing hallucinations and improving learning outcomes.

Key Benefits

- Accuracy: All responses grounded in course specific materials.
- Trust: Confidence aware mechanisms prevent hallucinated or unsupported answers.
- Personalization: Adaptive response modes (direct answer, hint, clarification).
- Integration: Easily embedded into LMS platforms (Moodle, Blackboard, Canvas).
- Cost Efficiency: Optimized RAG pipeline reduces inference and operational costs.

Why Institutions Will Adopt It -

- Enhances student performance and academic confidence
- Reduces lecturer workload by automating routine explanations
- Strengthens digital learning infrastructure

- Provides a competitive advantage in student support services

6.3 Revenue Model

1. Subscription Model (SaaS)
    - Universities pay a monthly or annual fee based on student count
    - Includes updates, maintenance, and technical support
2. Licensing Model
    - One-time license for institutions requiring on-premise deployment
    - Optional customization and integration services
3. API-Based Usage Model
    - EdTech companies pay per API call or per 1,000 queries
    - Suitable for integration into third-party learning platforms
4. Freemium Model (Optional)
    - Basic features free for students
    - Premium features (analytics, multi-course grounding) available for a fee

6.4 Cost Estimation

Development Costs

- Model hosting and compute
- Vector database infrastructure
- Backend/API development
- Testing and evaluation

Operational Costs

- Cloud hosting
- Model inference costs
- Maintenance and updates
- Customer support

Estimated Cost Breakdown

Cost Category	Estimated Cost (LKR)
Cloud compute & storage	50,000 – 80,000 / month
Development & maintenance	30,000 – 50,000 / month
Licensing & tools	10,000 – 20,000 / month
Support & monitoring	15,000 / month

Table. 3- Cost Estimation Table of component Hallucination – controlled, confidence-Aware LLM Tutoring Module

6.5 Pricing Strategy

Tiered Pricing

- Basic Tier: Limited queries, single-course grounding
- Standard Tier: Multi-course support, adaptive responses
- Premium Tier: Full analytics, institution-wide deployment

Competitive Pricing

- Priced lower than global EdTech tools (Khanmigo, Chegg AI Tutor)
- Affordable for Sri Lankan and regional markets

6.6 Competitive Advantage

Your system stands out due to:

1. Confidence-Aware Response Control -

Most AI tutors answer even when unsure.

Your system evaluates uncertainty and avoids hallucinations.

2. Course-Specific Grounding -

Responses are tied to actual lecture materials, not generic internet knowledge.

3. Ethical and Transparent AI -

Low-confidence responses trigger clarification instead of misinformation.

4. Modular Integration -

Easily integrates into LMS platforms and existing university systems.

5. Local Market Fit -

Designed for Sri Lankan and regional education systems, unlike global tools.

6.7 Intellectual Property Considerations

- The adaptive response mechanism, confidence-aware logic, and grounding enforcement pipeline can be patented as a novel AI tutoring framework.
- Course materials remain the property of the university; only embeddings are stored.
- Licensing agreements ensure institutions retain control over their academic content.
- The system complies with responsible AI guidelines and academic integrity standards.

7. BUDGET AND JUSTIFICATION

This section outlines the estimated costs required to design, develop, test, and evaluate the hallucination-controlled, confidence-aware LLM tutoring system. The budget includes hardware, software, cloud services, and human effort, with clear justification for each expense

7.1 Hardware Costs

Item	Estimated Cost (LKR)	Justification
Development Laptop / Workstation	150,000 – 200,000	Required for local experimentation, embedding generation, vector database setup, and backend development. A GPU-enabled machine significantly accelerates model testing and preprocessing.
External Storage (SSD)	15,000 – 25,000	Needed to securely store course materials, embeddings, logs, and experimental datasets during development.

Table. 4- Hardware Costs of component Hallucination – controlled, confidence-Aware LLM Tutoring Module

Justification

Hardware is essential for running local experiments, preprocessing uploaded academic materials, generating embeddings, and testing the RAG pipeline before cloud deployment. GPU-enabled hardware reduces development time and improves model performance during prototyping.

7.2 Software Licenses

Software / Tool	Estimated Cost (LKR)	Justification
IDE Tools (VS Code, PyCharm Community)	Free	Open-source tools sufficient for backend and API development.

Software / Tool	Estimated Cost (LKR)	Justification
Hugging Face Model Access	Free – 10,000	Many models are free; premium models may require subscription for higher-quality embeddings or inference.
Vector Database (FAISS / ChromaDB)	Free	Open-source and suitable for academic RAG pipelines.
API Testing Tools (Postman)	Free	Required for testing and validating backend endpoints.

Table. 5- Software Licenses of component Hallucination – controlled, confidence-Aware LLM Tutoring Module

Justification

Most AI development tools are open-source, reducing overall cost. Optional premium model access may be required for improved embedding quality or LLM inference performance.

7.3 Cloud Services

Service	Estimated Cost (LKR)	Justification
Cloud Compute (AWS / Azure / GCP)	40,000 – 70,000 / month	Required for hosting the LLM inference API, vector database, and backend services.
Storage (S3 / Blob Storage)	5,000 – 10,000 / month	Needed to store uploaded course materials, embeddings, and system logs.
Model Inference Costs	10,000 – 20,000 / month	Covers inference charges for generating responses using the LLM.

Table. 6- Cloud services of of component Hallucination – controlled, confidence-Aware LLM Tutoring Module

Justification

Cloud services ensure scalability, reliability, and availability. They allow the system to handle multiple users, run inference efficiently, and store academic materials securely. Cloud deployment is essential for real-world use by universities and EdTech platforms.

7.4 Human Effort Cost

Role	Estimated Cost (LKR)	Justification
AI Engineer / Researcher	50,000 – 80,000	Responsible for designing the RAG pipeline, uncertainty estimation algorithms, and adaptive response logic.
Backend Developer	30,000 – 50,000	Implements APIs, system integration, and backend architecture.
QA / Tester	20,000 – 30,000	Tests hallucination rate, accuracy, grounding quality, and system reliability.

Table. 7- Human effort cost of component Hallucination – controlled, confidence-Aware LLM Tutoring Module

Justification

Human effort is required for system design, implementation, evaluation, and documentation. Skilled personnel ensure the solution meets academic, ethical, and technical standards.

7.5 Justification Summary

Why These Costs Are Necessary

- Hardware enables efficient local development, preprocessing, and experimentation.
- Software tools provide access to high-quality models and development environments.
- Cloud services ensure scalable, secure hosting for the LLM tutor.
- Human effort ensures proper design, testing, and validation of the system.

Why the Budget Is Reasonable

- Most development tools are open-source, reducing cost.
- Cloud usage is optimized through an efficient RAG pipeline and selective inference.
- Human effort costs reflect realistic academic development requirements.
- The system is designed to be scalable and cost-efficient for long-term deployment.

8. WORK BREAKDOWN STRUCTURE (WBS)

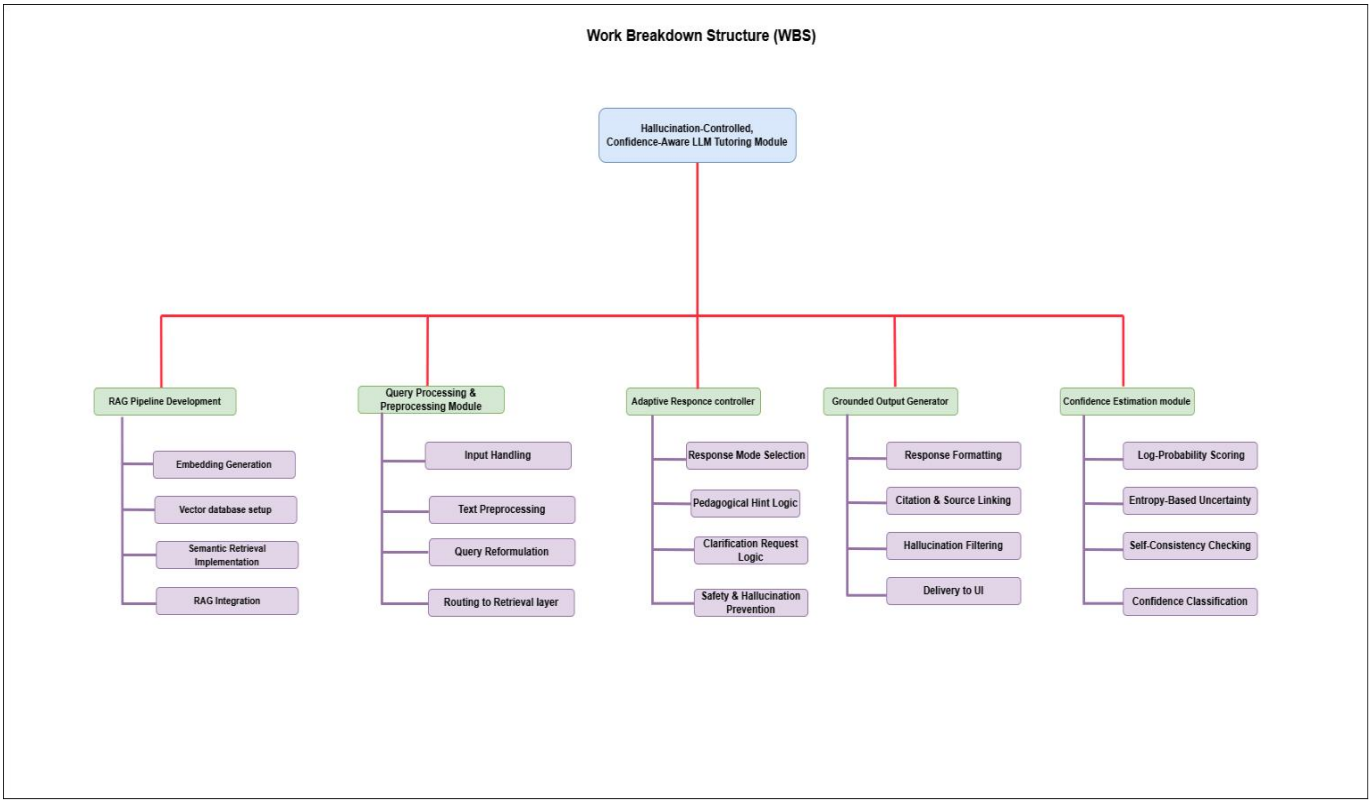


Figure 3- Work Breakdown of component Hallucination – controlled, confidence-Aware LLM Tutoring Module

9. GANTT CHART



Figure 4- Grantt chart of component Hallucination – controlled, confidence-Aware LLM Tutoring Module

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11. APPENDICES

11.1 Risk Analysis Summary

The Hallucination-Controlled, Confidence-Aware LLM Tutoring Module introduces several technical and ethical risks due to its reliance on Retrieval-Augmented Generation, uncertainty estimation, and adaptive response logic. The most significant risks arise from retrieval failures, inaccurate confidence estimation, and ungrounded LLM outputs, all of which can lead to hallucinated or misleading explanations. Additional risks include performance bottlenecks in the vector database, unclear hint generation, and potential ethical concerns if the system hides uncertainty from students. These risks are mitigated through multi-metric uncertainty estimation, strict grounding enforcement, threshold-based adaptive response control, and iterative evaluation with student feedback. Overall, the risk profile is manageable and aligns with the research-oriented nature of the component.

Sub-Component	Key Risks	Risk Level	Reason
RAG Pipeline Development	Poor retrieval quality, irrelevant documents, embedding mismatch	High	Retrieval errors directly cause hallucinations
Query Processing & Preprocessing	Incorrect intent detection, malformed queries	Medium	Can reduce retrieval accuracy but easier to fi
Confidence Estimation Module	Wrong confidence score, failure to detect hallucinations	High	Core research risk — uncertainty estimation is complex
Adaptive Response Controller	Wrong mode selection (answer/hint/clarification)	Medium	Impacts pedagogy but mitigated by thresholds
Grounded Output Generator	Missing citations, ungrounded content slipping through	High	Directly affects reliability and academic trust

Table 8- Risk analysis of the individual component